

JEE MAIN REVISION MASTER: ELECTROSTATICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 53 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Coulomb's Law, Electric Field & Dipole	44 PYQs	Forces & Fields
High	Gauss's Law & Electrostatic Potential	14 PYQs	Flux & Potential

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Coulomb's Law, Electric Field & Dipole	$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$, $E_{\text{axial}} = \frac{2kp}{r^3}$, $E_{\text{equatorial}} = \frac{kp}{r^3}$, $\vec{\tau} = \vec{p} \times \vec{E}$, $U = -\vec{p} \cdot \vec{E}$
Gauss's Law & Electrostatic Potential	$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$, $E = \frac{\sigma}{2\epsilon_0}$ (sheet), $E = \frac{\lambda}{2\pi\epsilon_0 r}$ (wire), $\vec{E} = -\nabla V$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Coulomb's Law, Electric Field & Dipole (Very High Yield — 44 PYQs)

Core Principle: Electrostatic forces, electric field intensity due to point charges and dipoles, and dipole torque/energy in uniform fields.

■ **Caution / JEE Pitfall:** $E_{\text{axial}} = 2E_{\text{equatorial}}$ for a short dipole. $\vec{E}_{\text{equatorial}}$ points opposite to dipole moment $\vec{p}$.

■ **Speed Trick:** Work done in rotating dipole from θ_1 to θ_2 : $W = pE(\cos\theta_1 - \cos\theta_2)$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "An electron falls through a distance 1.5 cm in 2.0×10^4 N/C from rest. The time taken to cover this distance in second is ____ ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "An electric dipole of dipole moment $\vec{p}$ is placed in the uniform electric field $\vec{E}$. Then which of the following statements are..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Charges are uniformly spread on the surface of a conducting sphere. The electric field from the centre of sphere to a point outside the sphere..." [Ans: N/A]

3.2 Gauss's Law & Electrostatic Potential (High Yield — 14 PYQs)

Core Principle: Gauss's flux theorem, symmetry applications (infinite sheet, cylinder, sphere), and electric potential gradient $\vec{\nabla}V = -\vec{E}$.

■ **Caution / JEE Pitfall:** Inside a charged hollow conducting sphere, electric field is ZERO, but potential is NON-ZERO and constant: $V_{\text{(inside)}} = (kq / R)$.

■ **Speed Trick:** Electric field lines are always perpendicular to equipotential surfaces and point in the direction of steepest potential decrease.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A 200 J of work is done in moving a charge 5 C from a point A where the potential is -20 V to another point B where potential is V volt..." [Ans: N/A]
- **[2026 Shift] Q#3 (MCQ):** "Consider three point charges $-2Q, Q$ and $-Q$ and three surfaces S_1, S_2 and S_3 as shown in the figure. Match the entries of List..." [Ans: N/A]
- **[2026 Shift] Q#4 (MCQ):** "What will be the total electric flux through the faces of the cube as given in the figure with side of length 'a' if a charge Q is placed..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 200 J of work is done in moving a charge 5 C from a point A where the potential is -20 V to another point B where potential is V volt. The value of V at B is

- (A) 10 V
- (B) 20 V
- (C) 40 V
- (D) 60 V

Correct Answer: None

Solution Analysis: Work done in moving a charge from one point to another is related to potential difference by $W = q(V_B - V_A)$. Here, $W = 200\text{ J}$, $q = 5\text{ C}$, $V_A = -20\text{ V}$. So, $200 = 5[V_B - (-20)]$ $200 = 5(V_B + 20)$. Divide both sides by 5 : $40 = V_B + 20$ $V_B = 20\text{ V}$. So, the value of potential at point B is 20V Correct option: Option B.

Benchmark Problem 2: KCET 2026 (Q#1)

An electron falls through a distance 1.5 cm in $2.0 \times 10^4\text{ N/C}$ from rest. The time taken to cover this distance in second is ____ ($e = 1.6 \times 10^{-19}\text{ C}$, $m_e = 9.11 \times 10^{-31}\text{ kg}$)

- (A) 2.9×10^{-9}
- (B) 2.9×10^9
- (C) 4×10^{-6}
- (D) 4×10^6

Correct Answer: None

Solution Analysis: The electron starts from rest, so its initial velocity is $u = 0$. The electric force on the electron is $F = eE$. Using Newton's second law, $F = ma$ so the acceleration is $a = \frac{eE}{m_e}$. Substitute the given values: $a = \frac{(1.6 \times 10^{-19})(2.0 \times 10^4)}{9.11 \times 10^{-31}}$ $a \approx 3.51 \times 10^{15}\text{ m/s}^2$. The distance covered is $s = 1.5\text{ cm} = 1.5 \times 10^{-2}\text{ m}$. Since the electron starts from rest, $s = \frac{1}{2}at^2$. So, $t = \sqrt{\frac{2s}{a}}$ $t = \sqrt{\frac{2(1.5 \times 10^{-2})}{3.51 \times 10^{15}}}$